UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2007 question paper

4024 MATHEMATICS

4024/01 Paper 1, maximum raw mark 80

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UNIVERSITY of CAMBRIDGE
International Examinations

1	**(a)**	15½ or 15.5	1		Not 31/2
	(b)	(0).175	1	2	
2	**(a)**	$\frac{11}{28}$	1		If answer decimal in range 0.39 to 0.40, accept fraction in working
	(b)	10	1	2	
3		(d =) 12 (implied by 8/12)	1		Accept answer reversed without evidence After 0+0.allow M1 for $\frac{8\times 3}{2}$ or $\frac{39\times 2}{3}$ seen M1
		(n =) 26 (implied by 26/39)	1	2	
4	**(a)**	(0).6 or $\frac{3k}{5k}$ (m/s²)	1		Accept – 0.6 etc
	(b)	72 (km/h)	1	2	
5	**(a)**	$\frac{66}{100}$ $\frac{666}{1000}$ 0.6r 0.67 **or** 0.6 $\frac{66}{100}$ $\frac{666}{1000}$ 0.67	1		Accept any equivalents
	(b)	1.507×10^{9} cao	1	2	
6	**(a)**	2 × 7 × 11 (× 1)	1		Accept without "×" if clear (e.g. 2, 7, 11)
	(b)	1078 cao	1	2	
				12	**12**
7	**(a)**	36	1		Accept 36°
	(b)	B + C = 180 or A + D = 180	1	2	Accept reference to interior or allied angles supplementary

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8	**(a)**	R correctly marked	1		2 squares "below" P
	(b)	$-\frac{3}{4}$ or – (0).75	1	2	
9	**(a)**	*y* marked in correct region	1		Within A ∩ B′. Not just shading
	(b)	**(i)** 6	1		Without brackets or braces
		(ii) 4 , 5	1	3	Ignore extra braces
10	**(a)**	(– 1 , 3)	1		
	(b)	$y < 3$ oe	1		Accept ≤ for < etc in both cases
		$y > \frac{1}{2}x$ oe	1	3	Both reversed,after 0 + 0 allow C1
				10	**10**
11	**(a)**	**(i)** $p = 40$	1		
		(ii) $q = 18$	1		Accept answer(s) reversed without evidence
	(b)	Rectangle, width 30, height 0.4	1	3	
12	**(a)**	1	1		
	(b)	32	1		Accept ±32, but not – 32 alone
	(c)	25	1	3	
13	**(a)**	4	1		
	(b)	**(i)** 2	1		Accept 2/1 or 4/2
		(ii) 1.1 to 1.3	1	3	Ignore any value of *y*
				9	**9**

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14	**(a)**	$6\pi x$ (cm) only	1		Accept numerical π in both parts
	(b)	$3\pi x^2$ (cm²) or unsimplified equiv (e.g. $4\pi x^2 - \pi x^2$)	2	3	$3\pi x^2 / 2$ or unsimplified equiv C1 **or** area larger (semi)circle = $(½)\pi(2x)^2$ M1 **or** area smaller (semi)circle = $(½)\pi x^2$ M1
15	**(a)**	42 (cm²)	1		
	(b)	16 (cm)	2	3	Use of similar triangles e.g. $\frac{h-4}{h} = \frac{9}{12}$ M1 **or** use their BCDE = ½ 12(4 + *h*) - ½ 9*h*
16	**(a)**	2/3	1		Accept 0.666 or better
	(b)	$\frac{3x+4}{5}$ asc	2	3	$ax + b$ with $a = 3/5$ $b \neq 0$ or $a \neq 0$ b=4/5 seen M1 Use of letter other than *x*, give - 1 if possible
17	**(a)**	(*DCB* =) 140	1		Not reflex angle
	(b)	(*DCF* =) 105 or {245 – their **(a)**} √	1		No √ for 122.5 then 122.5
	(c)	(*EFC* =) 75 or {180 – their **(b)**} √	1	3	**SC** 155, 90 , 90 scores 0, 1√, 0
18	**(a)**	($) 14 000	1		
	(b)	25 (%)	2	3	Answer 75 or 125 C1 figs $\frac{4.20-3.15}{4.20}$ oe seen M1
			15		**15**

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19 **(a)** $AY = BC$ (=3)
$CY = AB$ (=5)
AC common
Triangles congruent — 2
Conclusion must be stated
but SSS not needed unless extra facts are quoted
Two correct pairs of facts stated C1

(b) Area AYC = area ABC
(Add ACX)
$AYCX = ABCX$ — 1
Use of "right angle" scores 0

(c) Kite — 1 — 4

20 **(a)** 1 : 500 — 1
Accept 500

(b) **(i)** Triangle drawn with angles $35° \pm 2°$ and $70° \pm 2°$ — 1

(ii) 340 or 200 — 1
Accept $340° \pm 2°$ or $200° \pm 2°$

(iii) 28 to 31 (m) — 1 — 4
dep on scoring 1 in **(b)(i)**

8 **8**

21 Condone missing outside brackets, "= 0" and use of wrong letter if clear. If only "solutions" (even incorrect) in answer space, give marks if factors seen

(a) $(2x + 3)(x – 5)$ oe — 2
$(2x – 3)(x + 5)$ C1
or $\frac{7 \pm \sqrt{169}}{4}$ or better seen M1

(b) $(2y – z)(t – 4s)$ oe — 2 — 4
Complete correct extraction of one factor M1
such as $2yt – 8ys – z(t – 4s)$

22 **(a)** **(i)** $k > 2$ — 1
Accept $2 < k$ Not just 2

(ii) (t =) 30 — 1

(b) $x = 18½$ or 18.5
$y = 10½$ or 10.5 — 3 — 5
$x = \frac{37k}{2k}$ and $y = \frac{21k}{2k}$ seen C2
or one correct **with supporting working** C2
or one correct improper answer **with supporting working** C1
or correct method to eliminate x or y, M1
condoning 1 arithmetic slip

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23	**(a)**	1	1			
	(b)	**(i)** $x^2 - 4x = 0$ correctly obtained **AG**	2		$(3x + 1)^2 = (x + 1)^2 + (3x)^2$ oe seen M1 [Condone $3x^2$ for $(3x)^2$ for M1 etc]	
		(ii) $(x =)$ 4	1		Ignore $x = 0$	
		(iii) $-\frac{5}{13}$ or $-\frac{\text{their (ii)}+1}{3\,\text{their (ii)}+1}$ √	1	5	If answer decimal look back for fraction	
				14		**14**
24	**(a)**	$\begin{pmatrix}3\\7\\0\end{pmatrix}$	1			
	(b)	**(i)** **(a)** $(p =)$ 5	1			
		(b) $(w =)$ 4 or $14 - 2 \times$ their p √	1		Only allow √ if their $w > 0$	
		(ii) 7600	1		Ignore lack of braces	
		(iii) Total cost of making (100) buses and (200) lorries	1	5	Accept reasonable equivalents involving **cost** (and **toys)**	
25	**(a)**	**(i)** $125 < h \leq 135$ implied	1		Not just 20	
		(ii) 126.25, 126.2, 126.3, 126 or 126¼	3		**or** Correct method such as $\frac{5\times110+10\times120+20\times130+5\times140}{5+10+20+5}$ M2 condoning one error or omission **or** consistent use of wrong h in above M1 **or** $5 \times 110 + 10 \times 120 + 20 \times 130 + 5 \times 140$ M1	
	(b)	**(i)** 11 (cm)	2		121 or 132 used	
		(ii) 16	1	7		
				12		**12**

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